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ORIGINAL ARTICLE

KCNJ5-Negative Primary Aldosteronism: Histopathology and Surgical Outcomes

Che-Hsiung Wu, Kang-Yung Peng, Ya-Wen Yang, Peng-Ying Chen, Ya-Li Chen, Chin-Chen Chang, Yen-Hung Lin, Vin-Cent Wu; on behalf of the TAIPAI Study Group

BACKGROUND: Unilateral primary aldosteronism (uPA) is the most common surgically curable secondary hypertension. *KCNJ5* mutations are frequent in aldosterone-producing adenomas, but the clinical significance of *KCNJ5*-negative uPA remains unclear. We investigated whether histopathologic subtypes defined by the HISTALDO (Histopathology of Primary Aldosteronism) consensus are associated with postoperative outcomes in *KCNJ5*-negative uPA.

METHODS: We analyzed 171 *KCNJ5*-negative uPA patients from the Taiwan Primary Aldosteronism Investigation registry who underwent adrenalectomy. Somatic mutations were identified by Sanger sequencing and targeted next-generation sequencing. Adrenal histopathology was classified as classical (solitary adenoma or dominant nodule) or nonclassical (multiple nodules, micronodules, or diffuse hyperplasia) per the HISTALDO criteria. Postsurgical outcomes at 12 months were assessed using the Primary Aldosteronism Surgical Outcome consensus criteria.

RESULTS: The cohort (mean age, 55.0±11.3 years; 53.8% women) comprised 90 (52.6%) patients with classical histopathology and 81 (47.4%) with nonclassical features. Complete clinical success was achieved in 40.9% of patients at 12 months. Patients with classical histopathology had significantly higher complete clinical success (50%) than those with nonclassical histopathology (30.9%; P=0.01). On multivariate analysis, nonclassical histopathology (odds ratio, 0.33 [95% CI, 0.16–0.68]; P=0.003), higher preoperative serum creatinine (odds ratio, 0.24; P=0.022), and higher diastolic blood pressure (odds ratio, 0.97; P=0.017) independently predicted failure to achieve complete clinical success.

CONCLUSIONS: Nearly half of *KCNJ5*-negative uPA cases exhibit nonclassical adrenal histopathology, which is associated with significantly lower complete clinical success rates. These findings underscore the prognostic value of histopathologic classification in *KCNJ5*-negative uPA and support integrating histology and genetics to improve patient management. ***(Hypertension.* 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.26266.)** • **Supplement Material**.

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Key Words: adrenalectomy ■ blood pressure ■ hypertension ■ mutation ■ prognosis

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Primary aldosteronism (PA), characterized by autonomous excessive aldosterone production, accounts for 5% to 20% of hypertension cases in specialized clinics.[1,2] Among its subtypes, unilateral PA (uPA) caused by an aldosterone-producing adenoma (APA) is the most common form of secondary hypertension that can be corrected surgically. Morphological examination of resected adrenals, especially when combined with CYP11B2 (aldosterone synthase) immunohistochemistry, has revealed a spectrum of histopathologic patterns in uPA. The HISTALDO (Histopathology of Primary Aldosteronism) consensus was recently established to standardize terminology and classification of these adrenal lesions.[3] Early studies using the HISTALDO criteria reported that nonclassical histopathology (such as multiple APMs or diffuse hyperplasia) is associated with a higher likelihood of persistent aldosteronism after surgery compared with classical single-adenoma presentations.[4]

Research into the mechanisms of aldosterone overproduction in PA has largely focused on identifying somatic mutations. *KCNJ5* potassium channel mutations are the most prevalent genetic drivers of APAs, especially

Correspondence to: Vin-Cent Wu, MD, PhD, Department of Internal Medicine, National Taiwan University Hospital, Room 1419, Clinical Research Building, 7 Chung-Shan S Rd, Taipei 100, Taiwan. Email q91421028@ntu.edu.tw
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NOVELTY AND RELEVANCE

What Is New?

This is the first study to specifically investigate the impact of HISTALDO (Histopathology of Primary Aldosteronism) histopathologic subtypes on surgical outcomes in a large cohort of *KCNJ5*-negative patients with primary aldosteronism. We identified a high prevalence of nonclassical histopathology in this subgroup and demonstrated its independent association with lower complete clinical success rates.

What Is Relevant?

Understanding why some patients fail to achieve complete clinical success after adrenalectomy is a major clinical challenge. Our findings identify nonclassical adrenal pathology as a key predictor of poorer outcomes in the difficult-to-treat *KCNJ5*-negative population.

Clinical/Pathophysiological Implications?

KCNJ5-negative primary aldosteronism shows marked histological heterogeneity, with 47% of cases exhibiting nonclassical features that independently predict lower complete clinical success rates despite high complete biochemical success rates. Histopathology is a critical prognostic factor beyond genotype, enabling precise risk stratification. Nonclassical cases require counseling regarding continued antihypertensive therapy, intensified metabolic monitoring, and early intervention before irreversible organ damage develops.

Nonstandard Abbreviations and Acronyms

APA	aldosterone-producing adenoma
APM	aldosterone-producing micronodule
APN	aldosterone-producing nodule
AVS	adrenal venous sampling
DBP	diastolic blood pressure
DDD	Defined Daily Dose
HISTALDO	Histopathology of Primary Aldosteronism
NGS	next-generation sequencing
OR	odds ratio
PA	primary aldosteronism
uPA	unilateral primary aldosteronism

in East Asian populations (occurring in ≈50% to 70% of APAs versus 20% to 40% in Western cohorts).[5,6] By contrast, patients lacking *KCNJ5* mutations present a more heterogeneous clinical challenge. This *KCNJ5*-negative group is driven by a diverse array of less common somatic mutations in genes such as *ATP1A1*, *ATP2B3*, *CACNA1D*, *CACNA1H*, *CLCN2*, and *CTNNB1*. Non-*KCNJ5* APAs have been reported to more frequently consist of compact, eosinophilic, zona glomerulosa-like cells and often manifest as multiple small nodules rather than a single dominant tumor, in contrast to the lipid-rich, solitary tumors typical of *KCNJ5*-mutant APAs.[7] Consequently, patients with *KCNJ5*-negative uPA generally exhibit more variable—and often poorer—clinical outcomes after adrenalectomy compared with their *KCNJ5*-positive counterparts.[8,9] For example, *KCNJ5*-negative APA patients tend to have less pronounced hyperaldosteronism at baseline but may experience lower rates of hypertension remission and require prolonged follow-up for metabolic complications such as diabetes mellitus[10] and for cardiovascular risk reduction.[11,12] While advanced techniques (combining CYP11B2 immunohistochemistry with next-generation sequencing [NGS]) can identify a driver mutation in up to 85% of APAs, the clinical utility of detailed genotyping in the *KCNJ5*-negative subset remains uncertain.

Although *KCNJ5*-negative uPA is increasingly recognized as a distinct clinical entity, key knowledge gaps persist. The range and prevalence of histopathologic subtypes in this group—and their prognostic relevance—have not been fully delineated. It is also unclear how histological heterogeneity (eg, a classical APA coexisting with multiple micronodules) might impact surgical outcomes in *KCNJ5*-negative cases. In light of these gaps, we conducted a comprehensive study of adrenal histopathology and postsurgical outcomes specifically in *KCNJ5*-negative uPA, with parallel genotyping of known APA-driver genes. Our objectives were to determine (1) the prevalence of classical versus nonclassical histopathologic forms in *KCNJ5*-negative uPA, (2) the relationship of these histopathologic subtypes to surgical outcomes (clinical and biochemical success), and (3) the spectrum of aldosterone driver gene variants present in *KCNJ5*-negative uPA, particularly in those with classical histopathology.

MATERIALS AND METHODS

Study Population

The data that support the findings of this study are available from the corresponding author upon reasonable request. This study cohort was derived from the Taiwan Primary Aldosteronism

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Investigation (TAIPAI) prospective database and tissue bank.[13,14] uPA patients evaluated at National Taiwan University Hospital and collaborating centers (2 medical centers, 3 affiliated hospitals, and 2 regional hospitals across Taiwan) were enrolled between September 2016 and December 2021.[15,16] All patients had confirmed PA based on an elevated aldosterone-renin ratio and at least 1 positive confirmatory test (such as saline infusion or captopril challenge[17]). To avoid interference with testing, patients' antihypertensive medications were withdrawn for ≥21 days before confirmatory testing; if needed, diltiazem and doxazosin were used to control blood pressure during this washout period.[18,19] Adrenal lateralization (unilateral versus bilateral aldosterone excess) was determined via adrenal venous sampling (AVS) or NP-59 scintigraphy in all surgical candidates (Figure S1).[20]

Among 171 *KCNJ5*-negative patients, AVS was attempted in 119 (69.6%) cases. Successful bilateral adrenal vein cannulation, confirmed by a selectivity index (ratio of adrenal vein cortisol to peripheral vein cortisol) ≥2.0 bilaterally, was achieved in 107/119 (89.9%) patients.[21] In these successfully cannulated cases, unilateral aldosterone hypersecretion was diagnosed by a lateralization index (ratio of aldosterone-to-cortisol on the dominant side to that on the contralateral side) ≥2.0.[15] The remaining 12 patients with unsuccessful or unilateral AVS cannulation underwent complementary dexamethasone-suppressed 131I-6β-iodomethyl-19-norcholesterol (NP-59) SPECT/CT imaging for lateralization confirmation.[22] For patients in whom AVS was not attempted (n=52), lateralization was established by NP-59 SPECT/CT.[23] Thus, of 171 *KCNJ5*-negative patients, 107 were lateralized by AVS and 64 by NP-59 (52 primary + 12 complementary; Figure 2A). Detailed imaging criteria for lateralization are provided in the Supplemental Methods. Patients with confirmed unilateral PA proceeded to laparoscopic adrenalectomy of the affected gland, performed by experienced surgeons via a lateral transperitoneal approach. Resected adrenal specimens were immediately processed for pathology and also frozen at −80°C for research. Postoperatively, patients were followed closely (monthly for the first 3 months, then quarterly up to 1 year and beyond). The National Taiwan University Hospital Research Ethics Committee approved the study (No. 200611031R), and all patients provided written informed consent.

Histopathology and Terminology (HISTALDO Consensus)

All adrenal specimens underwent standardized pathological examination. Formalin-fixed, paraffin-embedded sections were stained with hematoxylin and eosin. Adjacent sections were processed for immunohistochemistry using monoclonal antibodies against CYP11B2 (aldosterone synthase) and CYP11B1 (11β-hydroxylase) to distinguish aldosterone-producing cell clusters. Staining was performed with a Novolink polymer detection system (Leica Biosystems) following the manufacturer's protocol. An Olympus BX51 microscope and DP72 camera were used for imaging. Each *KCNJ5*-negative case was classified according to the HISTALDO consensus criteria[3] using standardized terminology. Detailed definitions for specific histopathologic entities—including APA, aldosterone-producing nodule (APN), multiple APNs, aldosterone-producing micronodules (APMs), and aldosterone-producing diffuse hyperplasia—are provided in the Supplemental Methods. Based on these definitions, patients were categorized into 2 groups: the Classical Group (solitary APA or APN) and the Nonclassical Group (multiple APNs, APMs, or aldosterone-producing diffuse hyperplasia without a single dominant mass).[4,15,16]

Mutation Analysis

We performed targeted genetic analysis of each adrenal tumor to identify somatic mutations in major aldosterone-driving genes. Genomic DNA was extracted from CYP11B2-positive tumor regions (identified by immunohistochemistry) using microdissection and a QIAamp DNA Mini Kit (Qiagen, Hilden, Germany). All cases with solitary APAs or dominant nodules underwent CYP11B2-guided microdissection of the dominant lesion. We first conducted Sanger sequencing for hotspot mutations in *KCNJ5*, *ATP1A1, ATP2B3, CACNA1D*, and *CTNNB1* in tumors that presented as a solitary APA or a dominant APN. Samples without identifiable mutations by Sanger sequencing were subsequently analyzed with targeted NGS. A custom amplicon-based NGS panel was designed to cover coding exons (and flanking intronic sequences) of 7 genes commonly implicated in PA: *KCNJ5*, *ATP1A1, ATP2B3, CACNA1D, CACNA1H, CLCN2*, and *CTNNB1*.[24] Library preparation utilized a multiplex PCR-based approach on a Fluidigm Access Array system,[25] combining 48 samples and 48 primer pools per run.[26] Amplicons (up to ≈300 bp each) were barcoded and sequenced on an Illumina MiniSeq platform. For any mutation-positive tumor, matched leukocyte DNA from the same patient was sequenced to confirm the somatic nature of the mutation.

Genotyping Strategy for Nonclassical Histopathology

Given the technical challenges associated with extracting sufficient high-quality DNA from microscopic cell clusters, we adopted a tiered genotyping strategy based on lesion morphology:

(1) Dominant Lesions: For multiple APN cases, all macroscopically distinct CYP11B2-positive nodules (macronodules) were individually microdissected. Genotyping was successfully performed on the dominant lesions in a representative subset of 32 nonclassical cases (≈39.5%). In all cases, sequencing confirmed the absence of *KCNJ5* mutations.

(2) Validation of Adjacent Micronodules: For the microscopic background (APM or aldosterone-producing diffuse hyperplasia), which is morphologically indistinguishable from the adjacent zona glomerulosa on hematoxylin and eosin staining, comprehensive genotyping of every single focus was not feasible due to limited DNA yield. To rigorously validate the absence of *KCNJ5* mutations in this background, we performed a specific validation study on an additional 20 nonclassical cases. We used a pooled microdissection strategy. All anatomically corresponding CYP11B2-positive foci across 6 to 8 serial sections of the adjacent cortex were microdissected and pooled to ensure sufficient DNA yield. Sanger sequencing of these pooled samples confirmed the absence of *KCNJ5* mutations (representative examples are shown in Figure S3). Detailed

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protocols, including primer sequences and PCR conditions, are provided in the Supplemental Methods.

Outcomes of Interest

Follow-Up Protocol and Outcome Assessment

Patients were evaluated monthly for the first 3 months after adrenalectomy, then quarterly thereafter up to 12 months and beyond. A minimum follow-up of 6 months was required for inclusion in outcome analysis. Final clinical and biochemical outcomes were assessed at the 12-month postoperative visit, consistent with the Primary Aldosteronism Surgical Outcome consensus criteria[27] interval of 6 to 12 months. When a patient's classification differed between earlier visits and the 12-month assessment, the 12-month outcome was used as definitive to capture the sustained treatment effect.

Medication Burden Analysis

Medication burden was quantified using the World Health Organization Defined Daily Dose (DDD) methodology, comparing preoperative and 12-month postoperative antihypertensive and potassium supplement usage. The DDD is the assumed average maintenance dose per day for a drug used for its main indication in adults. Total DDD was calculated for each patient by summing the DDDs of all antihypertensive agents and potassium supplements.

Statistical Analysis

Statistical analyses were performed using R (version 4.0, R Foundation for Statistical Computing, Vienna, Austria). Continuous variables describing baseline patient characteristics are presented as mean ± SD if normally distributed and were compared using the Student *t* test. Nonnormally distributed continuous data are summarized as medians (interquartile range) and were compared using the Mann-Whitney *U* test. For paired pre- to postoperative comparisons, we additionally report the mean within-patient change with its 95% CI to provide complete inferential context. Categorical variables were compared using the χ^2 test or Fisher exact test as appropriate. We used logistic regression to identify predictors of complete clinical success. First, univariate analysis was performed for each potential predictor (demographic, clinical, and pathology variables). Variables with $P<0.05$ in univariate analysis, as well as other clinically important factors, were entered into a multivariable logistic regression model to determine independent predictors. To avoid multicollinearity between systolic and DBP (which are highly correlated), only diastolic blood pressure (DBP) was included in the multivariable model, as DBP showed a stronger univariate association with the outcome and greater pathophysiological relevance to aldosterone-mediated vascular resistance. Variance inflation factors were calculated for all predictors in the final model to assess multicollinearity. Odds ratios (ORs) with 95% CIs were calculated.

The performance of the multivariable logistic regression model was assessed through multiple metrics. We calculated the events-per-variable ratio to ensure it exceeded the recommended threshold of 10, thereby minimizing the risk of overfitting.[28] Model discrimination was quantified using the C statistic (area under the ROC curve), while calibration was evaluated using the Hosmer-Lemeshow goodness-of-fit test and visual inspection of calibration plots. To estimate optimism and obtain optimism-corrected performance metrics, internal validation was performed using bootstrap resampling with 300 iterations.

Sensitivity analyses were conducted to verify the robustness of our findings: (1) A prespecified reduced model including only 3 key predictors (baseline creatinine, DBP, and histopathology) was fit to assess robustness to the variable selection method; (2) Firth penalized logistic regression was performed as an additional check against separation bias; and (3) a subgroup analysis restricted to the 107 patients with successful AVS was performed to confirm that the inclusion of NP-59–lateralized cases did not confound the primary outcome associations.

Finally, to assess the robustness of significant associations against unmeasured confounders, we calculated *E*-values for the odds ratios of key predictors. The *E*-value represents the minimum strength of association (on the risk ratio scale) that an unmeasured confounder would need to have with both the exposure and the outcome to fully explain away the observed association. Calculations were performed using the formula derived by VanderWeele and Ding[29]: *E*-value = *RR* + RR×*(RR-1), where *RR* is the risk ratio (or the reciprocal of the odds ratio for protective estimates). These computations were verified using the EValue package in R. Throughout the analysis, a 2-tailed $P<0.05$ was considered statistically significant.

RESULTS

Prevalence and Genetic Profiling of *KCNJ5*-Negative uPA

Of the 295 uPA patients who underwent adrenalectomy, 124 (42.0%) had somatic *KCNJ5* mutations in their tumors, whereas 171 (58.0%) were confirmed as *KCNJ5*-negative uPA. This 42% overall prevalence encompasses all histopathologic subtypes, including nonclassical cases in which *KCNJ5* mutations are inherently less common. When restricted to patients with classical histopathology (solitary adenomas or dominant nodules, n=214), the *KCNJ5* mutation prevalence was 58% (124/214; Figure S2). The mean age of *KCNJ5*-negative patients was 55.0±11.3 years, and 92 (53.8%) were female. Among the 90 *KCNJ5*-negative patients with classical histopathology, initial Sanger sequencing of DNA from CYP11B2-positive tumor regions identified known driver mutations in several genes: *CTNNB1* in 14 cases (15.6%), *CACNA1D* in 6 (6.7%), *ATP1A1* in 4 (4.4%), and *ATP2B3* in 3 (3.3%). The remaining 63 tumors that were Sanger-negative underwent targeted NGS, which revealed additional somatic mutations in 37 of them: *CACNA1D* in 20 cases, *ATP1A1* in 6, *ATP2B3* in 3, *CACNA1H* in 5, *CTNNB1* in 2, and *CLCN2* in 1 case. No additional *KCNJ5* mutations were identified beyond the initial Sanger screening. Combining these results, the distribution of driver mutations in *KCNJ5*-negative classical APAs was dominated by *CACNA1D* (26 cases, 28.9%), followed by *CTNNB1* (16 cases, 17.8%), *ATP1A1* (10

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cases, 11.1%), *ATP2B3* (6 cases, 6.7%), *CACNA1H* (5 cases, 5.6%), and *CLCN2* (1 case, 1.1%). Despite comprehensive sequencing, 26 of the classical tumors (28.9%) remained mutation-negative for all tested genes. These proportions are illustrated in Figure 1. Overall, a somatic driver mutation was identified in 71.1% of classical *KCNJ5*-negative APAs, underscoring the considerable molecular diversity underlying aldosterone overproduction in this subgroup. For context, in the entire cohort of solitary APAs or dominant nodules, *KCNJ5* mutations remained the most prevalent (58%), followed by *CACNA1D* (12%) and *CTNNB1* (8%), with 12% lacking any known somatic mutation (Figure S2).

Clinicopathologic Characteristics by Histopathology

According to HISTALDO criteria, the 171 *KCNJ5*-negative cases were classified into 90 (52.6%) with classical histopathology and 81 (47.4%) with nonclassical histopathology. Figure 2A illustrates the classification of all adrenal cases in the cohort. Notably, even among the 90 classical cases, 59 (65.6%) had additional small aldosterone-producing lesions in the adjacent cortex, despite the presence of a single dominant APA/APN (Figures 2B and 3).

The classical and nonclassical groups were similar in age, sex distribution, duration of hypertension, and tumor size (Table 1). However, the classical group exhibited significantly higher systolic blood pressure at diagnosis, higher plasma aldosterone concentration, and higher aldosterone-renin ratio (Table 1). Among the 107 patients with successful AVS lateralization, key parameters indicating the degree of unilateral aldosterone hypersecretion were comparable between classical and nonclassical groups. The median lateralization index was 15.2 (IQR, 8.3–24.7) in classical cases versus 14.1 (IQR, 8.9–23.8) in nonclassical cases (P=0.67). Contralateral suppression ratios were similarly equivalent (P=0.46; Table S1). These findings confirm that both histopathologic subtypes had similarly robust biochemical lateralization.

Histopathology and Postsurgical Outcomes

Qualitatively, the adrenal tumors in *KCNJ5*-negative uPA were typically smaller and composed of cells resembling those of the zona glomerulosa (Figure 3). They tended to grow in well-defined nests, cords, or trabeculae, whereas *KCNJ5*-mutated adenomas often displayed a solid, sheet-like architecture. Notably, some *KCNJ5*-negative tumors (particularly those with *CACNA1D* mutations) contained prominent myxoid (connective tissue) stroma, whereas *KCNJ5*-mutated tumors more frequently showed cellular or nuclear atypia.

Antihypertensive medication burden decreased significantly following adrenalectomy in the overall *KCNJ5*-negative cohort (mean DDD decreased from 2.03±1.36 at baseline to 1.26±1.52 at 12 months post-operatively; mean within-patient reduction 0.77 [95% CI, 0.51–1.03]; P<0.001 by paired Wilcoxon signed-rank test; Table S2). The large marginal standard deviations reflect substantial between-patient heterogeneity in baseline antihypertensive regimens, whereas the narrow 95% CI of the mean change confirms the consistency and statistical robustness of the treatment effect. The reduction was more pronounced in the classical group (preoperative 2.10±1.49 DDD to postoperative 1.14±1.73 DDD) compared with the nonclassical group (preoperative 1.94±1.18 to postoperative 1.40±1.18). Potassium supplementation decreased dramatically in both groups (mean DDD decreased from 0.57±0.18 at baseline to 0.04±0.16 at 12 months post-operatively;

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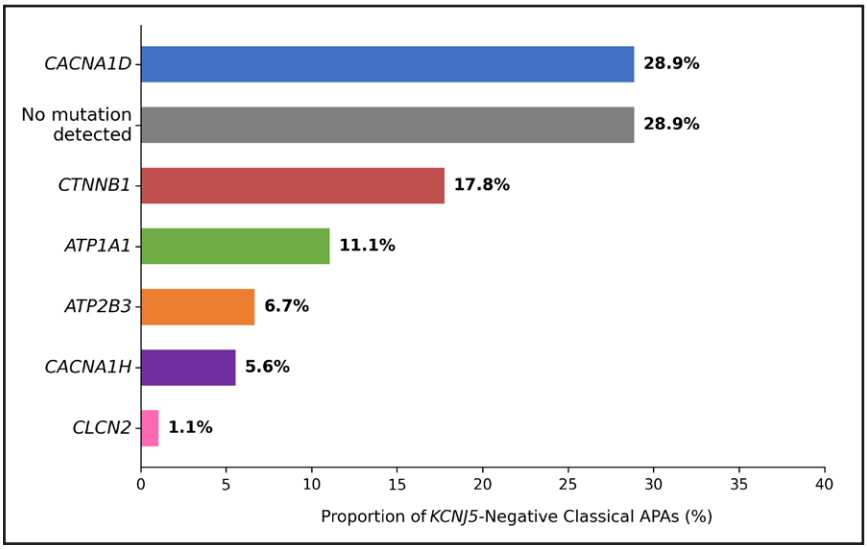

Figure 1. Prevalence of identified gene variants in solitary aldosterone-producing adenomas or dominant aldosterone-producing nodules without *KCNJ5* mutation.
The horizontal bar chart illustrates that *CACNA1D* mutations constitute the most prevalent driver (28.9%) in *KCNJ5*-negative classical cases, followed by *CTNNB1* (17.8%), *ATP1A1* (11.1%), *ATP2B3* (6.7%), *CACNA1H* (5.6%), and *CLCN2* (1.1%). 28.9% of cases remain mutation-negative despite comprehensive next-generation sequencing (NGS) analysis.

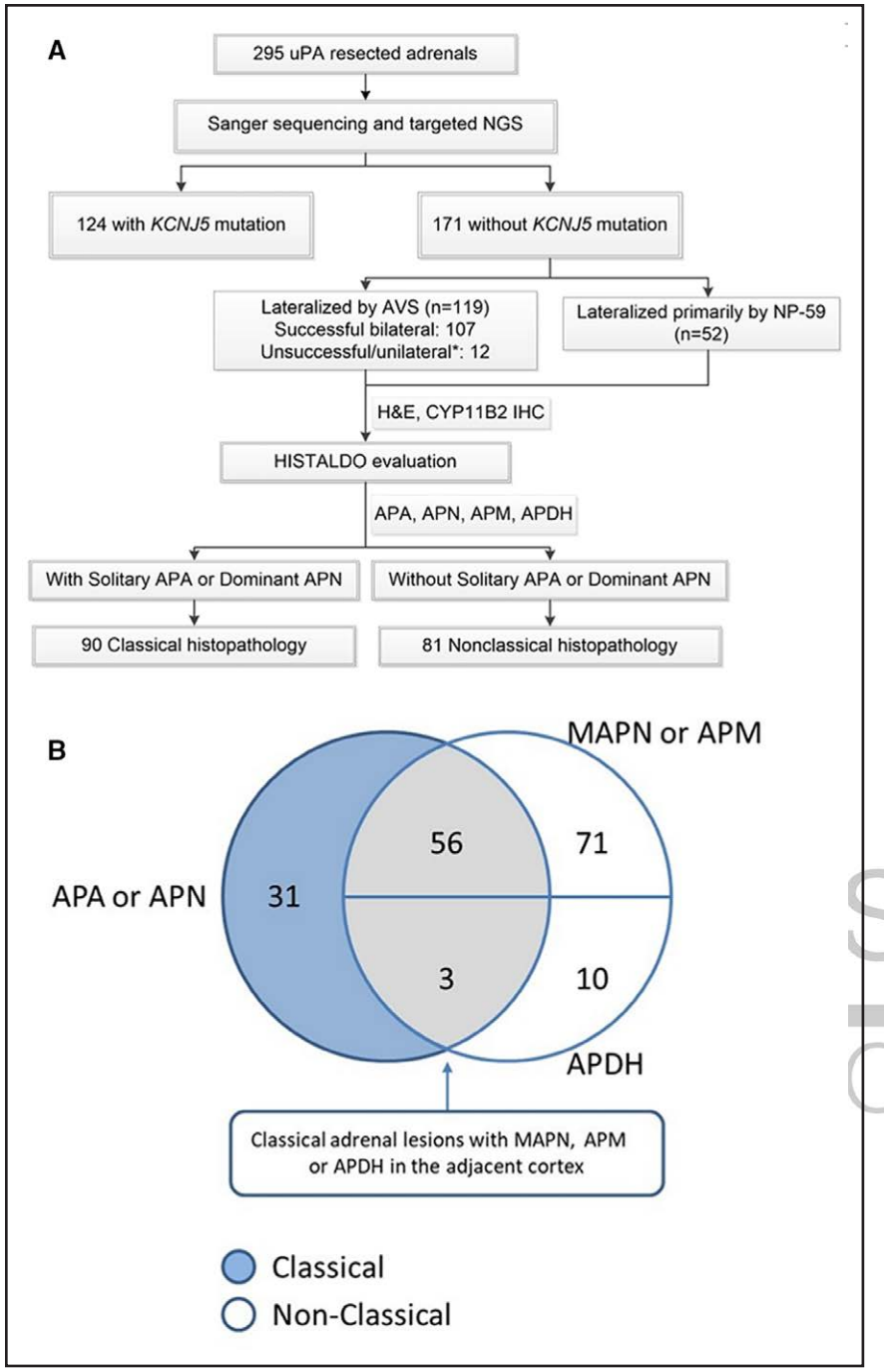

Figure 2. Histopathologic evaluation of adrenals according to histopathology of HISTALDO (Histopathology of Primary Aldosteronism) consensus. **A**, Study cohort flow diagram detailing the selection and classification of 295 unilateral primary aldosteronism (uPA) patients. Among 171 *KCNJ5*-negative patients (study cohort), preoperative lateralization was achieved by adrenal venous sampling (AVS) in 107 cases following successful bilateral cannulation (89.9% cannulation success rate among 119 AVS attempts), and by NP-59 SPECT/CT in 64 cases (52 as primary modality, 12 complementary to unsuccessful AVS*). All adrenals underwent hematoxylin and eosin (H&E) staining and CYP11B2 (aldosterone synthase) immunohistochemistry for histopathologic classification per HISTALDO consensus, yielding 90 classical and 81 nonclassical cases. Classical refers to a solitary aldosterone-producing adenoma (APA) or a dominant aldosterone-producing nodule (APN), whereas nonclassical includes multiple aldosterone-producing nodules (MAPN), micronodules (APM), or aldosterone-producing diffuse hyperplasia (APDH). *Footnote explains patients with unsuccessful AVS received complementary NP-59 for lateralization confirmation. **B**, Venn diagram demonstrating the distribution of histopathologic evaluation in *KCNJ5*-Negative uPA. In approximately two-thirds (59/90) of classical cases, small aldosterone-producing foci were also present in the adjacent cortex. Conversely, the nonclassical group exhibited MAPN, APM, or APDH.

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mean within-patient reduction 0.53 [95% CI, 0.50–0.56]; $P<0.001$ by paired Wilcoxon signed-rank test), with most patients discontinuing supplementation.

In terms of clinical outcomes, complete clinical success was achieved in 70 out of 171 (40.9%) patients in the overall *KCNJ5*-negative cohort (Table 1). This rate differed markedly by histopathology subtype: 50.0% (45/90) of patients with classical histopathology achieved complete clinical success, compared with only 30.9% (25/81) of those with nonclassical histopathology ($P=0.01$; Figure 4). Subgroup analysis within the classical histopathology cohort revealed that the presence of coexisting APMs (n=59) did not significantly influence surgical outcomes. When comparing classical cases with versus without adjacent micronodules, similar rates of complete clinical success were observed (52.5% versus 45.2% [31/59 versus 14/31], $P=0.659$) and complete biochemical success (84.7% versus 77.4% [50/59 versus 24/31], $P=0.365$). This indicates that the presence of ancillary APMs does not materially affect the therapeutic benefit of adrenalectomy in *KCNJ5*-negative classical cases (Table S3). Biochemical success rates were uniformly high and did not differ between classical and nonclassical groups (complete or partial biochemical success in >90% of patients

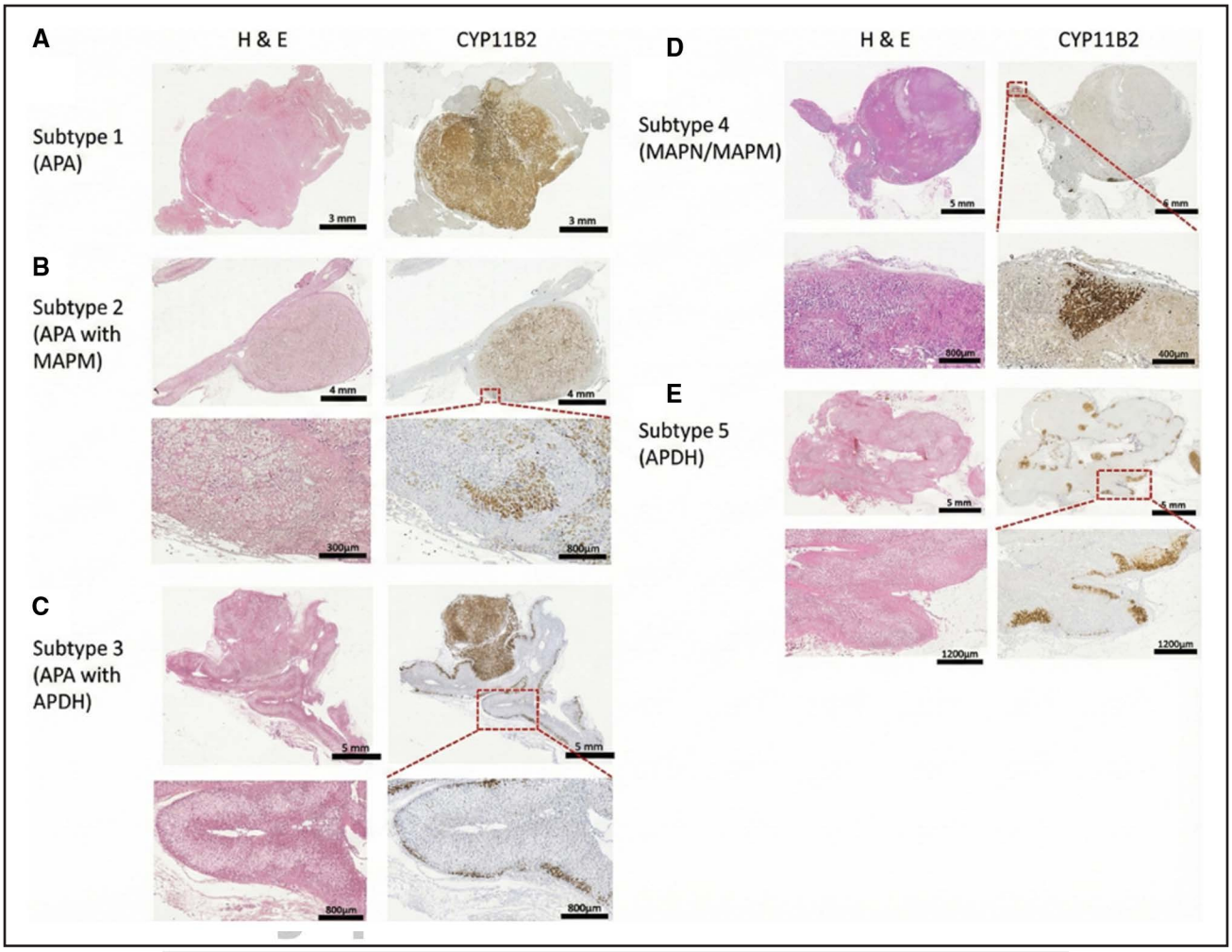

Figure 3. Representative histopathologic subtypes of *KCNJ5*-negative unilateral primary aldosteronism, defined by the pattern of hematoxylin and eosin (H&E) morphology and CYP11B2 immunohistochemistry, in accordance with the international HISTALDO (Histopathology of Primary Aldosteronism) consensus.

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Resected adrenal specimens were categorized into 5 histopathologic subtypes based on the architectural pattern of aldosterone-producing lesions: (**A**) subtype 1—aldosterone-producing adenoma (APA): a solitary aldosterone-producing adenoma with no CYP11B2-positive foci in the adjacent cortex; (**B**) subtype 2—APA with aldosterone-producing micronodule (APM): an APA accompanied by multiple aldosterone-producing micronodules (APMs) in the adjacent cortex, seen as discrete peritumoral CYP11B2-positive foci; (**C**) subtype 3—APA with aldosterone-producing diffuse hyperplasia (APDH): an APA coexisting with APDH in the surrounding cortex, showing a broad zone of subcapsular CYP11B2 positivity; (**D**) subtype 4—multiple aldosterone-producing nodule (MAPN)/APM-predominant: nonclassical morphology with multifocal aldosterone-producing nodules (MAPNs) or APMs without a dominant APA, CYP11B2 staining reveals multiple scattered positive clusters across the cortex; (**E**) subtype 5—aldosterone-producing diffuse hyperplasia (APDH) only: diffuse aldosterone-producing hyperplasia in the absence of a distinct tumor mass, with broad subcapsular CYP11B2 positivity. Each panel shows corresponding H&E and CYP11B2-stained sections. Red boxes indicate regions enlarged below each subtype panel. Scale bars are provided as labeled on each image.

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in both groups; P=0.948; see Table 1 for details). In other words, nearly all patients experienced normalization or improvement of aldosterone excess after surgery, but the resolution of hypertension was significantly more likely in those with classical adrenal pathology.

Finally, we identified independent predictors of complete clinical success using logistic regression. On univariate analysis, several factors were associated with higher odds of complete clinical success, including lower body mass index, lower preoperative systolic and DBP, shorter hypertension duration, lower serum creatinine, and classical (versus nonclassical) histopathology. To avoid multicollinearity, systolic blood pressure was excluded from the multivariate model, and only DBP was retained. In the final model (Table 2), nonclassical histopathology remained a strong independent predictor of lower complete clinical success (OR, 0.33 [95% CI, 0.16–0.68]; P=0.003). In addition, elevated preoperative serum creatinine (OR 0.24 per mg/dL [95% CI, 0.07–0.84]; P=0.022) and higher baseline DBP (OR, 0.97 [95% CI, 0.94–0.99]; P=0.017) were significantly associated with a lower likelihood of success. All variance inflation factor

Table 1. Clinical Characteristics in *KCNJ5*-Negative Unilateral Primary Aldosteronism, Stratified by Histopathologic Findings (Classical Versus Nonclassical)

Variable	Total cohort	Classical*	Nonclassical	*P* value
No. of patients, %	171	90 (52.6%)	81 (47.3%)	
Female, n (%)	92 (53.8%)	43 (47.8%)	49 (60.5%)	0.124
Age, y	55.0±11.3	54.6±10.0	55.4±12.6	0.615
Body mass index, kg/m^2	25.8±4.3	26.0±4.3	25.7±4.2	0.655
Systolic BP, mm Hg	153.8±21.4	157.1±22.1	150±20.0	0.031
Diastolic BP, mm Hg	90.7±14	92.4±13.3	88.8±14.3	0.095
PRA, ng/mL per min†	0.8±2.5	0.7±2.0	1.0±3.0	0.373
Plasma aldosterone, ng/dL†	48.9±33.3	56.4±36.4	40.5±27.1	0.001
Log ARR†	2.2±0.8	2.4±0.7	2.0±0.8	0.008
Serum potassium, mmol/L†	3.4±0.8	3.4±0.7	3.4±0.8	0.981
Serum creatinine, mg/dL	0.94±0.39	0.93±0.31	0.95±0.46	0.858
Tumor size, cm	1.8±0.9	1.8±0.8	1.8±0.9	0.913
Duration of hypertension, y	8.3±8.2	8.6±8.0	8.0±8.4	0.672
baPWV, cm/s‡	1602 (1444–1860)	1622 (1445–1860)	1595 (1427–1853)	0.54
Clinical success,§ %				0.033
Complete	70 (40.9%)	45 (50%)	25 (30.9%)	
Partial	62 (36.3%)	29 (32.2%)	33 (40.7%)	
Absent	39(22.8%)	16(17.8%)	23(28.4%)	
Biochemical success,§ %				0.948
Complete	142 (83%)	74 (82.2%)	68 (84%)	
Partial	13 (7.6%)	7 (7.8%)	6 (7.4%)	
Absent	16 (9.4%)	9 (10%)	7 (8.6%)	

Data are expressed as mean ± SD unless otherwise indicated. ARR indicates aldosterone-renin ratio; baPWV, brachial-ankle pulse wave velocity; BP, blood pressure; and PRA, plasma renin activity.

*Among classical cases, 59 (65.6%) had additional aldosterone-producing lesions in the adjacent cortex. Detailed outcome stratification by micronodule presence is provided in Table S3.

†Obtained after holding drugs that interfere with the renin-angiotensin system.

‡Expressed as median (interquartile range).

§Definition of postoperative outcome: Evaluation was performed at 12 mo post-operatively according to the Primary Aldosteronism Surgical Outcome consensus. Complete clinical success was defined as normal blood pressure without antihypertensive medication. Complete biochemical success was defined as correction of hypokalemia and normalization of the aldosterone-renin ratio based on local laboratory reference intervals.

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values in the final model were <2 (range, 1.12–1.42), indicating no significant multicollinearity (Table S4).

Model Performance and Validation

The final multivariable model was based on 70 complete clinical success events and 5 predictors, resulting in an events-per-variable ratio of 14.0, which exceeds the recommended threshold for model stability. The model demonstrated acceptable discrimination with an apparent C statistic of 0.737 (95% CI, 0.658–0.816). Calibration was adequate, as indicated by a nonsignificant Hosmer-Lemeshow test (χ^2=13.06, df = 8, *P*=0.110). We further assessed model performance using a calibration plot (Figure S4), which showed that predicted probabilities generally aligned with observed outcome frequencies along the diagonal, confirming that the model is well calibrated and reliable for risk estimation. To quantify potential overfitting, internal validation using bootstrap resampling (300 iterations) was performed. This revealed a small mean optimism of 0.032, yielding an optimism-corrected C statistic of 0.707 (95% CI, 0.625–0.789) and a calibration slope of 0.85, suggesting the model maintains good discrimination when applied to new patients.

Sensitivity analyses confirmed the robustness of the final model across different modeling assumptions and subgroups. First, in the prespecified reduced model (creatinine, DBP, histopathology), nonclassical histopathology remained a highly significant independent predictor with a virtually identical effect size (OR, 0.31 [95% CI, 0.15–0.65]; *P*=0.002) and a comparable C statistic (0.709). Second, Firth penalized logistic regression yielded consistent results (OR, 0.34 [95% CI, 0.16–0.70]; *P*=0.004), ruling out separation bias (Table S5). Third, in the subgroup restricted to 107 patients with successful AVS, nonclassical histopathology remained a highly significant predictor of lower complete clinical success (adjusted OR, 0.18 [95% CI, 0.05–0.62]; *P*=0.006), showing an even stronger

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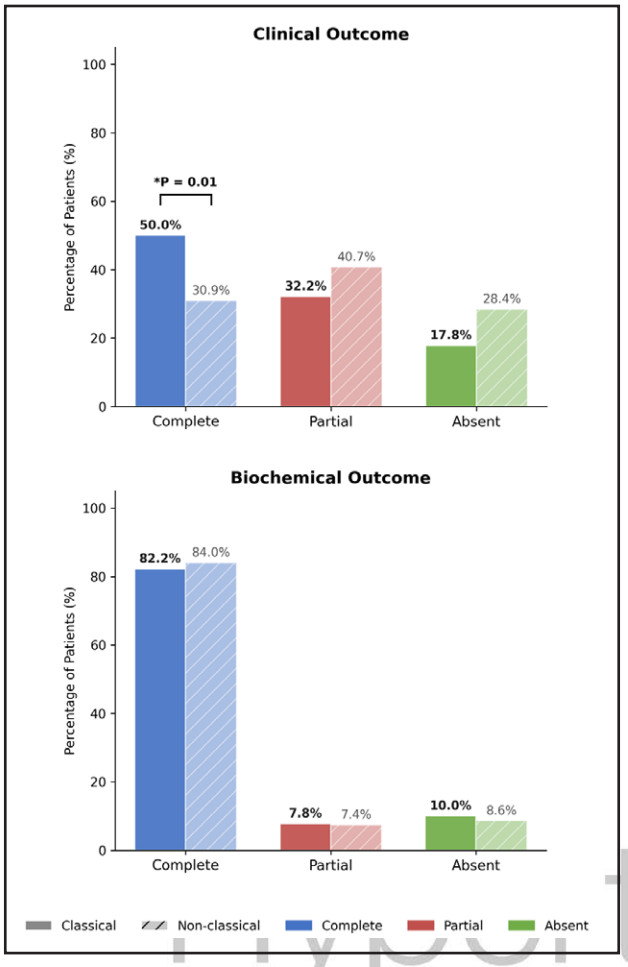

Figure 4. Postsurgical outcomes stratified by histopathologic findings according to the HISTALDO (Histopathology of Primary Aldosteronism) consensus in Unilateral Primary Aldosteronism Without *KCNJ5* Mutations. Grouped vertical bars display the proportion of patients achieving complete, partial, or absent success for clinical outcomes (**left**) and biochemical outcomes (**right**) at 12 months post-adrenalectomy, evaluated per the Primary Aldosteronism Surgical Outcome (PASO) consensus criteria. Solid bars, classical group; hatched bars, nonclassical group. *P=0.01 for the comparison of complete clinical success between groups. Biochemical success rates were comparable between groups (P=0.948).

association than in the full cohort (Table S6). Finally, the E-value for nonclassical histopathology was 5.4 (confidence limit 2.3), indicating that an unmeasured confounder would need an implausibly large association (risk ratio >5.4) to fully explain away the observed results.

DISCUSSION

To our knowledge, this is the first study to specifically delineate the interplay between adrenal histopathology, somatic mutation profile, and surgical outcomes in the *KCNJ5*-negative PA population. Three key findings emerge from our data. First, the overall rate of complete clinical success (blood pressure normalization without medications) in *KCNJ5*-negative uPA was 40.9%, which is considerably lower than the complete clinical success rates often reported for *KCNJ5*-mutant uPA. Second, nearly half (47.4%) of *KCNJ5*-negative cases exhibited nonclassical histopathology, and this subgroup had a significantly lower complete clinical success rate (about 31%) compared with those with classical single-adenoma pathology (50%). This histological dichotomy—classical versus nonclassical adrenal architecture—clearly emerged as a key factor associated with postoperative outcome in patients without *KCNJ5* mutations. For patients with *KCNJ5*-negative PA, biochemical remission represents the minimal surgical end point; however, histopathologic subtype is the principal association with long-term blood pressure benefit. Third, somewhat surprisingly, the presence of small aldosterone-producing foci adjacent to an APA (which we observed in two-thirds of classical cases) did not adversely impact the chances of complete clinical success in *KCNJ5*-negative patients. Taken together, these findings establish histopathologic subtype as a critical prognostic factor in *KCNJ5*-negative uPA and suggest that this subgroup of PA represents a distinct clinical entity requiring a tailored management approach.

Histopathologic Complexity and Postsurgical Outcomes

One of the most striking characteristics of *KCNJ5*-negative uPA in our study was its marked histopathologic heterogeneity. The prevalence of nonclassical histology (multiple nodules or diffuse hyperplasia) reached 47.4% in our cohort—nearly double the ≈25% rate reported in mixed-genotype PA cohorts that include *KCNJ5*-positive cases.[4,16] Importantly, we found that this higher prevalence of nonclassical features in *KCNJ5*-negative patients correlated with worse surgical outcomes, evidenced by a ≈19% absolute difference in complete clinical success rates between the classical and nonclassical groups. Multivariate analysis confirmed that histopathology remains an independent predictor of outcome even after accounting for other factors. This association appears robust against potential confounding (E-value = 5.4), reinforcing the link between adrenal histological pattern and surgical outcome.

Intriguingly, this histological impact was most evident in the dissociation between biochemical and clinical outcomes. While biochemical success was achieved at similarly high rates in both classical and nonclassical groups, complete clinical success was significantly less frequent in the latter (50.0% versus 30.9%). This divergence likely reflects the distinct determinants of these end points. Biochemical success after adrenalectomy is primarily a technical outcome reflecting the accurate

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Table 2. Predictors of Complete Clinical Success After Adrenalectomy in *KCNJ5*-Negative Unilateral Primary Aldosteronism

	Univariate		Multivariate*	
Variable	Odds ratio (95% CI)	*P* value	Odds ratio (95% CI)	*P* value
Sex (female)	0.59 (0.32–1.10)	0.097		
Age, y	0.98 (0.96–1.01)	0.183		
BMI, kg/m^2	0.93 (0.86–0.99)	0.041	0.94 (0.87–1.02)	0.14
SBP, mm Hg	0.98 (0.96–0.99)	0.007		
DBP, mm Hg	0.97 (0.94–0.99)	0.006	0.97 (0.94–0.99)	0.017
PRA, ng/mL per min†	0.98 (0.86–1.12)	0.769		
Plasma aldosterone, ng/dL†	1.01 (1.00–1.01)	0.302		
Log ARR†	1.09 (0.74–1.60)	0.679		
Potassium, mmol/L†	0.96 (0.87–1.08)	0.244		
Serum creatinine, mg/dL	0.16 (0.05–0.54)	0.003	0.24 (0.07–0.84)	0.022
Duration of hypertension, y	0.95 (0.91–0.99)	0.03	0.96 (0.92–1.01)	0.083
Tumor size, cm	1.31 (0.92–1.86)	0.132		
Histopathology (nonclassical)	0.45 (0.24–0.84)	0.012	0.33 (0.16–0.68)	0.003

ARR indicates aldosterone-renin ratio; BMI, body mass index; BP, blood pressure; DBP, diastolic blood pressure; PRA, plasma renin activity; and SBP, systolic blood pressure.

*Model includes baseline diastolic BP (not systolic BP) to avoid multicollinearity in the multivariate analysis.

†Obtained after holding drugs that interfere with the renin-angiotensin system.

identification and complete removal of the dominant aldosterone source. In our cohort, rigorous lateralization followed by total adrenalectomy reliably accomplished this regardless of histopathologic subtype. This aligns with multicenter immunohistopathology studies demonstrating that while histological patterns (classical versus nonclassical) correlate strongly with clinical risk profiles, immediate biochemical cure remains high across subtypes provided lateralization is accurate.[30] Consistent with prior observations, *KCNJ5*-negative patients typically achieve biochemical cure but exhibit worse clinical outcomes, including higher metabolic risks.[10] Even in nonclassical cases with multifocal nodules, total adrenalectomy removes the entire dominant source; residual contralateral APMs typically produce insufficient aldosterone to meet PA diagnostic criteria postoperatively.[31]

In contrast, complete clinical success depends on multiple factors beyond source removal. First, clinical success is constrained by irreversible cardiovascular remodeling induced by long-standing aldosterone excess. As demonstrated by Rossi et al,[32] structural vascular changes such as arterial stiffening persist despite biochemical normalization.[33] Our multivariable analysis supports this mechanism, as higher preoperative DBP—a surrogate for increased peripheral vascular resistance and arteriolar remodeling—independently predicted significantly lower odds of clinical success. Patients with nonclassical histopathology may represent a more indolent, long-standing disease process, allowing greater cumulative vascular damage to accrue before diagnosis. Second, nonclassical histopathology—characterized by multifocal or diffuse aldosterone-producing lesions—may reflect a more extensive or asymmetrical bilateral adrenal involvement. Even after unilateral adrenalectomy, residual contralateral aldosterone production might persist, with subtle aldosterone activity remaining below the diagnostic threshold yet sufficient to sustain hypertension.[34,35] This dissociation underscores the importance of histopathologic subtype in predicting surgical outcomes beyond simple biochemical normalization after adrenalectomy.

Incidental Micronodules in Classical Cases Do Not Impair Outcomes

It is worth noting that histopathologic complexity is not exclusive to the nonclassical group. Even within the classical *KCNJ5*-negative cases, we observed that 65.6% had some degree of adjacent aldosterone-producing lesions in the cortex. This rate is higher than the ≈51% reported in series enriched with *KCNJ5*-mutant cases,[16] suggesting that even histologically solitary tumors in *KCNJ5*-negative PA often exist alongside a more complex adrenal microarchitecture. Adjacent APMs (historically termed aldosterone-producing cell clusters[3]), which frequently harbor *CACNA1D* mutations, are known to accumulate with age.[36,37] In the present study, we observed that having a few additional micronodules around a dominant tumor did not compromise the complete clinical success rate, which remained much higher than in the nonclassical group. This observation is consistent with recent work by Ahn et al,[38] who similarly categorized classical uPA into solitary versus mixed subtypes based on the presence of coexisting microlesions. Interestingly, they found that while solitary and mixed classical cases differed in some biochemical measures (with solitary APAs showing the highest aldosterone

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levels and lowest potassium), both had similarly favorable clinical outcomes postsurgery, and both achieved better blood pressure control with fewer medications than the nonclassical group. All patients in our cohort underwent total adrenalectomy once unilateral lateralization was confirmed. This is crucial because preoperative imaging and AVS often cannot distinguish between a pure solitary APA and a mixed pattern. By removing the entire affected gland, we ensure the complete excision of both the dominant tumor and any coexisting micronodules. Consequently, our data confirm that as long as the dominant driver is removed via total adrenalectomy, the incidental presence of secondary microlesions does not compromise the therapeutic benefit.

Genotype-Phenotype Interplay Beyond KCNJ5

The differences between *KCNJ5*-mutant and non–*KCNJ5* APAs, both histologically and clinically, are well documented. *KCNJ5* mutations lead to APAs composed of lipid-laden, zona fasciculate-like cells that typically form a solitary, larger nodule, whereas *KCNJ5*-wildtype (nonmutated) tumors are more often small, zona glomerulosa-like and can present as multiple nodules .[39,40] However, genotype alone is not an absolute predictor of outcome. The observed complete clinical success rate of 40.9% in our *KCNJ5*-negative cohort warrants contextualization within this framework. Several factors may explain this favorable outcome despite the absence of *KCNJ5* mutations, which are traditionally associated with better prognosis. Our cohort possessed several prognostically favorable baseline characteristics, including female predominance, relatively younger age, and lower body mass index—all well-established predictors of superior blood pressure outcomes following adrenalectomy.[41] Our 40.9% complete clinical success rate should also be evaluated within the broader context of international outcomes. Zhou et al[42] reported a pooled complete clinical success rate of 50.6% (95% CI, 43%–58%) across multiple studies, with East Asian cohorts demonstrating notably higher rates (61.3% in Chinese populations versus 43.7% in other ethnic groups). More specifically, Zhang et al[43] reported a complete clinical success rate of 35.2% in *KCNJ5*-wildtype patients using Primary Aldosteronism Surgical Outcome criteria. Our observed rate of 40.9% is comparable to this benchmark, suggesting that while *KCNJ5*-negative patients have lower success rates than their *KCNJ5*-mutant counterparts, a substantial proportion still achieve biochemical and clinical cure. Furthermore, at the tissue level, differences in CYP11B2 (aldosterone synthase) expression between *KCNJ5*-mutant and nonmutant APAs are not as dramatic as one might expect—quantitative image analysis has shown no significant difference in CYP11B2 staining intensity by genotype.[40] These complexities suggest that an integrative approach is needed: incorporating genotypic information with detailed histopathologic assessment may yield the most accurate risk stratification for patients. Our data support this, as both a genetic factor (indirectly, *KCNJ5*-negative status correlating with the presence of alternative mutations) and a histological factor (HISTALDO classification) together helped explain the variation in surgical outcomes. Ultimately, understanding how specific mutations (such as *CACNA1D* or *CTNNB1*) correlate with certain histological patterns and clinical phenotypes could pave the way for personalized prognosis in PA.

Landscape of Genetic Diversity and Unresolved Questions

Our findings on the genetic landscape of *KCNJ5*-negative classical APAs show striking concordance with recently published international data. For instance, Meyer et al[4] reported in a German cohort that among classical APAs, 56% had *KCNJ5* mutations, 10% had *CACNA1D*, and ≈15% had no mutation identified. In our cohort, we observed remarkably similar proportions among classical cases: 57.9% *KCNJ5*-positive (excluded from this analysis by definition), 12.1% *CACNA1D*-positive, and 12% mutation-negative. This strong agreement across different populations underscores the robustness of the histopathology-guided sequencing approach and suggests that the core genetic architecture of classical APAs may transcend ethnic differences.[44]

Focusing on the *KCNJ5*-negative subset, genetic profiling revealed several noteworthy findings. First, *CACNA1D* emerged as the predominant driver mutation in *KCNJ5*-negative classical APAs. Second, we observed a *CTNNB1* mutation prevalence of 17.8% in *KCNJ5*-negative classical APAs (8% when considering all classical cases including *KCNJ5*-positive), which is higher than the ≈5% reported in some European cohorts.[45] This likely reflects two factors: (1) comprehensive NGS detection: unlike previous large-scale studies that used Sanger sequencing targeting only canonical *CTNNB1* hotspot mutations (S33, S37, S45), our targeted NGS panel covered the entire coding sequence. This enabled the detection of noncanonical mutations, such as the p.Gln123Ter (Q123*) nonsense mutation identified in our cohort, which lies outside the traditional N-terminal hotspot region and would be missed by hotspot-only Sanger sequencing. Recent studies using comprehensive sequencing have similarly identified *CTNNB1* mutations at positions Asp32, Gly34, and others that were previously unrecognized.[46] (2) Population variation: just as *KCNJ5* (higher in East Asians) and *CACNA1D* (higher in Black populations) show ethnic variation, *CTNNB1* prevalence may also differ across populations.[47] Our predominantly Taiwanese cohort may harbor population-specific genetic or environmental modifiers that predispose to *CTNNB1*-mutant APAs.

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Despite comprehensive profiling, 28.9% of our *KCNJ5*-negative classical APAs lacked any known driver mutation, highlighting a significant gap in our understanding of PA pathogenesis. Ongoing research to discover new genetic drivers or epigenetic mechanisms in these "mutation-negative" cases is warranted.

Biological Rationale for *KCNJ5* Negativity in Nonclassical Lesions

We acknowledge that comprehensive genotyping of every microscopic focus is not feasible. However, extensive literature supports that *KCNJ5* mutations are exceptionally rare in APMs and nonclassical lesions. Nishimoto et al analyzed 23 APMs and found zero *KCNJ5* mutations (0/23).[48] Similarly, Omata et al[37] profiled 61 APMs from normal adrenals (0/61) and 99 APMs from idiopathic hyperaldosteronism (1/99),[34] finding *KCNJ5* mutations to be virtually absent (<1%) compared with *CACNA1D* mutations, with the single *KCNJ5*-mutant lesion being morphologically a micro-APA rather than a typical APM. Collectively, across these major studies profiling over 180 APMs/clusters, *KCNJ5* mutations were absent in >99% of cases.

A biologically plausible explanation for this scarcity relates to zonation-specific mutation tolerance. *KCNJ5* encodes an inwardly rectifying potassium channel that is physiologically enriched in zona glomerulosa cells and plays a key role in maintaining membrane hyperpolarization.[5,49] Gain-of-function *KCNJ5* mutations abolish potassium selectivity, resulting in sustained membrane depolarization and excessive calcium influx. In zona glomerulosa-like cells (characteristic of APMs), where baseline *KCNJ5* expression is high and calcium-buffering capacity is low, this mutation-induced calcium overload may exceed cellular tolerance thresholds, leading to early clonal loss rather than stable formation of micronodules. This negative selection pressure likely prevents the accumulation of *KCNJ5*-mutant clones in the nonclassical phase. In contrast, *KCNJ5*-mutant APAs frequently exhibit a zona fasciculata-like phenotype with clear-cell morphology and distinct transcriptional features.[50] Prior experimental studies suggest that the acquisition of a zona fasciculata-like cellular program and upregulation of calcium-buffering or antiapoptotic pathways (eg, VSNL1) may mitigate calcium-induced cytotoxicity, allowing the survival and subsequent clonal expansion of *KCNJ5*-mutant cells.[50,51] Accordingly, we speculate that viable *KCNJ5*-mutant clones may bypass the prolonged APM phase characteristic of nonclassical disease and instead preferentially progress toward zona fasciculata-like adenomatous lesions. While direct lineage-tracing evidence in humans remains limited, this hypothesis aligns with the observation that *KCNJ5* mutations are exclusive to classical APAs and absent in the nonclassical, multifocal phenotype.

Clinical Implications and Preoperative Risk Stratification

Several preoperative clinical parameters emerged from our study as useful indicators of outcome in *KCNJ5*-negative PA. Patients with classical histopathology tended to present with higher blood pressure, higher aldosterone levels, and higher aldosterone-renin ratios at baseline—consistent with harboring a single hyperactive aldosterone-secreting adenoma. These factors also translated to greater absolute improvements after surgery, as removal of the solitary source of aldosterone had a pronounced effect. In contrast, the nonclassical group's lower baseline aldosterone levels likely reflect a more distributed aldosterone production, which may explain the more modest blood pressure reduction from unilateral adrenalectomy. In addition, we identified elevated preoperative serum creatinine and higher DBP as independent predictors of poorer clinical outcomes after surgery. While elevated creatinine points to renal impairment,[52] higher DBP likely serves as a surrogate for increased systemic vascular resistance and fixed arteriolar remodeling—both hallmarks of chronic aldosterone toxicity. This suggests that patients who have sustained significant end-organ damage from long-standing hyperaldosteronism (or other comorbidities) are less likely to experience complete reversal of hypertension.[53] The association between creatinine and surgical outcome was particularly strong (OR ≈0.24 per 1 mg/dL; *E*-value 7.5), indicating that it is robust against confounding. From a practical standpoint, these findings reinforce the importance of early detection and treatment of PA—before significant irreversible damage (such as hypertensive nephropathy or vascular stiffening) occurs—to maximize the chance of achieving complete clinical success in hypertension. It also suggests that patients with kidney impairment or severe diastolic hypertension might need closer postoperative monitoring or adjunctive antihypertensive therapy even after adrenalectomy.

Limitations and Future Directions

Despite offering new insights into the histopathologic and genetic landscape of *KCNJ5*-negative PA, our study has several limitations. First, our cohort was drawn primarily from the TAIPAI registry, which is composed predominantly of East Asian patients. This may limit the generalizability of our findings to other ethnic groups or regions, especially given known variations in APA genotypes and clinical presentations across populations. Second, while our overall sample size was adequate to detect major differences, the numbers became small when stratifying by specific rare mutations. This limited our power to draw conclusions about genotype-phenotype correlations for those uncommon variants. Third, because we only included patients who underwent surgery, there is

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a selection bias—our results may not apply to all patients with PA (eg, those managed medically or those with bilateral disease). Additionally, our study design cannot establish causality; we observed associations (eg, between histology and outcomes), but functional or mechanistic studies are needed to prove causation and to unravel the underlying molecular pathways. Fourth, comprehensive genotyping was not performed on all nonclassical cases due to tissue constraints in extreme microscopic lesions. However, extensive literature supports the rarity of *KCNJ5* mutations in these lesions, and we validated this in a representative subset. Even if a small number of undetected *KCNJ5* mutations were present in the nongenotyped nonclassical group, this bias would actually strengthen our conclusions. Since *KCNJ5* mutations are typically associated with classical histology and better surgical outcomes, their inadvertent inclusion in the nonclassical group would bias the results toward the null (attenuating the observed difference). The fact that we still observe significantly poorer outcomes in the nonclassical group despite this potential conservative bias indicates that the association is robust. Finally, we cannot exclude the influence of unmeasured confounders such as dietary sodium intake, adherence to postsurgical lifestyle modifications, or other genetic factors affecting blood pressure.[54] However, the high *E*-value (5.4) obtained in our analysis suggests that the influence of unmeasured confounding would be minimal. Addressing these limitations will require further research. Ideally, multicenter prospective studies with larger and more ethnically diverse cohorts should be conducted to validate our findings. Such studies could also incorporate comprehensive omics profiling (genomics, transcriptomics) to identify novel markers of outcome in *KCNJ5*-negative PA. Ultimately, this line of research will help refine patient selection for surgery and follow-up strategies, ensuring that each patient with PA receives optimally tailored care.

Conclusions

Our study demonstrates that among patients with *KCNJ5*-negative uPA, histopathologic subtypes are strongly associated with surgical outcomes. Nonclassical histopathology was common in this group and was independently linked to lower rates of complete clinical success after adrenalectomy. Conversely, classical single-adenoma cases had higher complete clinical success rates, and the presence of small secondary APNs did not significantly affect outcomes. We also found a high prevalence of alternative somatic mutations in *KCNJ5*-negative APAs through comprehensive genetic profiling, highlighting the molecular heterogeneity underlying this condition. These findings support the integration of detailed histopathologic classification with genetic analysis to improve prognostic accuracy and guide personalized treatment in PA.

Perspectives

This study highlights the critical heterogeneity within *KCNJ5*-negative PA. We demonstrate that nonclassical histopathology is a frequent and distinct entity associated with poorer surgical outcomes. These findings suggest that the integration of histopathologic classification with genetic profiling is essential for refining prognosis. Future precision medicine strategies should consider these subtypes when counseling patients about the likelihood of complete clinical success and may explore whether patients with nonclassical features require more intensive postoperative management or monitoring for long-term recurrence.

ARTICLE INFORMATION

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Affiliations

Division of Nephrology, Taipei Tzu Chi Hospital, Buddhist Tzu Chi Medical Foundation, Taiwan (C.-H.W.). School of Medicine, Tzu Chi University, Hualien, Taiwan (C.-H.W.). Department of Physiology and Pharmacology, College of Medicine, Chang Gung University, Taoyuan, Taiwan (K.-Y.P.). Department of Surgery (Y.-W.Y.), Department of Internal Medicine (P.-Y.C., Y.-L.C.), and Division of Cardiology, Department of Internal Medicine (Y.-H.L.), National Taiwan University Hospital, Taiwan. Department of Medical Imaging, National Taiwan University Hospital, National Taiwan University College of Medicine, Taipei (C.-C.C.). Taiwan Primary Aldosteronism Center of Internal Medicine, National Taiwan University Hospital (PAC), Taiwan Primary Aldosteronism Investigators (TAIPAI), Taipei (V.-C.W.). Department and Graduate Institute of Forensic Medicine, National Taiwan University College of Medicine, Taipei (C.-C.C.).

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Disclosures

None.

Supplemental Material

Expanded Materials & Methods
Tables S1–S6
Figures S1–S4

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